

## Python Interview Prep Roadmap (Folder Structure)

### 01\_INTRODUCTION

- Installing Python
- Running Python scripts
- Python syntax
- Variables & data types
- Input/Output
- Basic operations

### 02\_CONTROL\_FLOW

- if / elif / else
- for loops
- while loops
- break, continue

### 03\_FUNCTIONS\_MODULES

- Creating functions
- Lambda
- Scope
- Importing modules

### 04\_DATA\_STRUCTURES

- Lists
- Tuples
- Dictionaries
- Sets

### 05\_STRINGS

- Indexing & slicing
- Joining & splitting
- Palindromes

## 06\_OBJECT\_ORIENTED\_PROGRAMMING

- Classes & objects
- Inheritance
- Magic methods

## 07\_ALGORITHMS\_AND\_DSA

07.1 Arrays & Lists

07.2 Strings

07.3 Hash Maps

07.4 Two Pointers

07.5 Stacks & Queues

07.6 Recursion

07.7 Sorting & Searching

07.8 Linked Lists

07.9 Trees

07.10 Graphs

## 08\_SQL\_FOR\_INTERVIEWS

- SELECT, WHERE, JOIN
- GROUP BY, ORDER BY

## 09\_API\_BUILDING

- Flask/FastAPI basics
- CRUD

- Connecting to DB

## 10\_INTERVIEW\_PRACTICE

- Warm-ups
- Easy/Medium/Hard problems

## 11\_SYSTEM\_DESIGN\_BASICS

- API design
- Caching
- Logging

## 12\_FINAL\_PRACTICE

- 50–100 LeetCode problems
- Review resume & GitHub